**SESSION 11 – NAIVE BAYES ALGORITHM**

🎯 **Topic**

- What is Naive Bayes
- Bayes Theorem (intuition only)
- Why "Naive"
- Types of Naive Bayes
- Real-life use cases
- Practical implementation

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## 1. 🎯 Learning Objectives

By the end of this session, students will be able to:

- Understand **Naive Bayes concept**
- Learn how probability is used in classification
- Understand the **independence assumption**
- Know different types of Naive Bayes
- Build a **classification model using Naive Bayes**
- Compare with Logistic Regression

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## 2. 🔁 Quick Recap (Connect Previous Session)

Start class like this:

"Till now, we used Logistic Regression for classification.
Today we will learn a different approach — one that is based completely on probability."

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## 3. 📖 What is Naive Bayes?

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### 3.1 Definition

Naive Bayes is a classification algorithm based on **probability and Bayes Theorem**.

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### 3.2 Simple Meaning (Most Important Line)

"Naive Bayes predicts class based on probability of features."

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## 4. 📊 Bayes Theorem (Concept Only)

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**Core Idea**

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

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**Simple Explanation**

“Probability of A given B depends on how likely B occurs when A is true.”

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**Teaching Line**

“We reverse probability to make predictions.”

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## 5. 📌 Why “Naive”?

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**Assumption**

All features are **independent** of each other.

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**Example**

Spam detection:

- Words in email are treated as independent

👉 In reality, not true → but still works well

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**Teaching Line**

“Naive Bayes assumes features don’t depend on each other.”

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## 6. 📊 How Naive Bayes Works

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**Step-by-Step**

1. Calculate probability of each class
2. Calculate probability of features given class
3. Multiply probabilities
4. Choose class with highest probability

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**Teaching Line**

“Naive Bayes selects the class with highest probability.”

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## 7. 📊 Real-Life Example

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**Example: Spam Detection**

Words: “Free”, “Offer”, “Win”

👉 Model calculates:

- P(Spam | Words)
- P(Not Spam | Words)

👉 Chooses higher probability

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## 8. 📊 Types of Naive Bayes

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🔹 **Gaussian Naive Bayes**

- Continuous data
- Example: height, marks

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🔹 **Multinomial Naive Bayes**

- Text data (word count)

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🔹 **Bernoulli Naive Bayes**

- Binary features (0/1)

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**Teaching Tip**

👉 For your class → focus on **Gaussian NB**

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## 9. 📊 Advantages

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✅ Very fast
✅ Works well on small data
✅ Good for text classification
✅ Simple to implement

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## 10. ⚠️ Limitations

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❌ Assumes independence
❌ Not always accurate for complex data

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### 11. 📊 Naive Bayes vs Logistic Regression

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| Feature | Naive Bayes | Logistic Regression |
|---|---|---|
| Based on | Probability | Optimization |
| Speed | Very fast | Moderate |
| Data assumption | Independent features | No strict assumption |

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### 12. 🎯 Interview Concepts

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**Q1: What is Naive Bayes?**

👉 Probability-based classification algorithm.

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**Q2: Why called Naive?**

👉 Assumes features are independent.

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**Q3: Where is it used?**

👉 Spam detection, text classification.

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**Q4: Which type for numeric data?**

👉 Gaussian Naive Bayes.

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### 13. 💻 PRACTICAL (YOUR FORMAT)

🎯 **Objective**

Classify student result using Naive Bayes

### 14. 📊 Output Interpretation

Explain:

- Model predicts using probability
- Faster than Logistic Regression
- Works well even with small dataset

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👉 Conclusion:

“Naive Bayes is simple but powerful for many real-world problems.”

“Naive Bayes is a simple yet powerful algorithm based on probability.
Even with its strong assumptions, it performs surprisingly well in many real-world applications, especially text classification.”